

**Response to Study by U.S. Copyright Office, Library of Congress on
“Role of Copyright Law with respect to software-enabled consumer products”**

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This document contains the response from Tata Consultancy Services (TCS) to the study initiated by U.S. Copyright Office, Library of Congress to review the role of copyright law with respect to software-enabled consumer products.

1. The provisions of the copyright law that are implicated by the ubiquity of copyrighted software in everyday products

Copyright is a legal right accorded by the law of a country (and extendable to other countries participating in the Berne convention) to grant the creator of an original work exclusive rights for its use and distribution for a limited defined period. In the US, the copyright law is founded on the constitutional goal to promote the progress of science and useful arts. Software copyright further protects this legal right for computer programs as a form of “literary work” and copyrighted software embedded in everyday products enjoys the same level of protection as that installed on general purpose computer.

The role of copyright law in enabling/frustrating the design, distribution, and legitimate uses of software-enabled consumer products and related services, as also its impact on business models for copyright owners and users is being debated strongly in the US and the world over. This is equally applicable to consumer devices with software embedded in them, but also to devices operated by software via a smartphone/remote controller or software housed on the cloud.

Consumer choice is key and US copyright law needs to ensure that consumers continue to have a wide variety of options while buying products/services than be locked into products/services of certain manufacturers/service providers. To ensure sustainable innovation in the area of software-enabled consumer product goods, it is essential that the competitive market landscape be strengthened by preventing business monopolies by companies in the name of copyright law.

It is our view that the US copyright law needs to balance between the rights of the creators of the software copyrighted work as well as recognize legitimate rights of consumers who buy software-enabled products/services from manufacturers, resellers/distributors and service providers. Also, companies are continuously innovating to bring out more cost-effective and advanced products/services into the market and may want to have the right to reverse engineer copyrighted software embedded in similar products from other companies, to make their software-enabled products/services interoperable with the software-enabled products/services of other companies.

While the ownership of the copyrighted software rests with the author(s) of such copyrighted work, copyright law also allows/would allow the owner of the program to make another copy or adaptation for use on a computer (17 U.S.C. 117) for indicative scenarios indicated below -

- i) Permitting the reproduction or adaptation of “the software or firmware of a user purchased mobile communications device for the sole purpose of . . . connect[ing] to a wireless communications network” if the reproduction or adaptation is initiated by or with the consent of the owner of the device, the owner is in legal possession of the device, and the owner has the consent of the authorized operator of the wireless communications network to use the network (proposed by the Unlocking Technology Act of 2015).
- ii) Allowing the transfer of ownership of a copy of a computer program embedded on a machine or other product “if [the] computer program enables any part of [that] machine or other product to operate,”

as well as any right to receive software updates or security patches from the manufacturer. This right of transfer could not be waived by any contractual agreement. In addition, the original owner of the device would be prohibited from retaining an unauthorized copy of the computer program after transferring the device and the computer program to another person. (proposed by You Own Devices Act).

Further, section 109 of the US Copyright Act prohibits the rental, lease or lending of a computer program for direct or indirect commercial gain unless authorized by the copyright owner of the program. It expressly provides an exception for “a computer program which is embodied in a machine or product and which cannot be copied during the ordinary operation or use of the machine or product.” and ensures that computer programs can be embedded in machines or products to avoid interference with the ordinary use of such products.

The above points indicate that US copyright law does have provisions in place to ensure rightful ownership of copyrighted software as well as enablement for various business scenarios involving software-enabled consumer products.

However, it is recommended that US copyright law (existing and proposed to be amended by respective acts indicated above) and related guidelines provide further clarity on issues related to reverse engineering (to create “transformative products”), to enable inter-operability etc. This will ensure that product manufacturers and service providers are clear on what the law enables/frustrates upfront without having to resort to expensive, protracted litigation and precedence from related case laws.

2. Whether, and to what extent, the design, distribution, and legitimate uses of products are being enabled and/or frustrated by the application of existing copyright law to software in everyday products

We referred some of the case laws involving the design, distribution and legitimate users of products containing copyrighted software. It is seen from these examples that copyright law enables these aspects of software-enabled consumer products. However, it was seen that in each case, the district court initially granted an injunction to the plaintiff citing a case of copyright infringement. When each case was appealed to the Federal circuit, the cases were decided in favor of the defendant, on grounds relating to fair use, merger doctrine, DMCA, first sale doctrine and other aspects that form an integral part of copyright law.

The asymmetry of information, market power and financial capacity in the context of such litigation by product owners and copyright holders may lead to a disincentive for consumers or third party innovators to defend their rights in every case to its logical end. It is therefore recommended to bring out further clarity in the US copyright law and related guidelines for all concerned stakeholders to minimize dependency on case laws for precedence in such cases.

i) In a specific instance, copying of a copyrighted software of a gaming products company (e.g. reverse engineering the BIOS) during the development of an emulator software by a competitor firm, first by using the unchanged software to develop emulation for the hardware, and then by developing a BIOS of their own does not constitute copyright infringement but is covered by fair use. This was especially so, as the final product of the competitor firm did not contain any infringing material. The court thereby held that the design of the competitor product was “moderately transformative” and permitted the distribution and legitimate use of the competitor product, thus enabled by the application of existing copyright law to software in everyday products. The application of copyright law in this context helped to ensure that the business does not enjoy a monopoly.

ii) In another instance, the court ruled that where reverse engineering is the “only means of gaining access to . . . unprotected aspects of the program, such copying is a fair use. The court judge the case based on the fair use provisions to decide whether the defending party had infringed on the code of the plaintiff. Considering that the defending party had to make a copy of the plaintiff’s code and de-compile it to

determine the password required to make its devices interoperable with that of the plaintiff's, the case was decided by the court in favor of the defendant.

In reaching its decision, the court noted that: [W]e are free to consider the public benefit resulting from a particular use notwithstanding the fact that the alleged infringer may gain commercially. Public benefit need not be direct or tangible, but may arise because the challenged use serves a public interest. It is precisely this growth in creative expression, based on the dissemination of other creative works and the unprotected ideas contained in those works, that the Copyright Act was intended to promote. This case also shows that copyright law is enabling the design, distribution and legitimate uses of software-enabled consumer products.

iii) In another instance, the court analyzed the case of circumvention of the plaintiff's authentication system both from the perspective of possible violation of the Digital Millennium Copyright Act (DMCA) and also taking into consideration the idea/expression dichotomy aspects. The court held that a program cannot be copyrightable based on the assumption that it could be written in a number of ways without considering the practical realities. Also, from the DMCA claims made by the plaintiff, the court ruled that since the authenticated sequence did not "control access" to the program which the defendant needed for interoperability purposes and that the purchase of the plaintiff's device itself allowed access to the program, it was decided that the defendant did not violate DMCA.

Since the concluding opinion was that even if the programs involved were more complex (and thus more deserving of copyright protection), the key question would be the purpose of the circumvention technology. If a third party manufacturer's use of a circumvention technology was intended only to allow its products to interoperate with another manufacturer's, and not to gain any independent benefit from the functionality of the code being copied, then that circumvention would be permissible.

The copyright law thus was interpreted by the court to serve as an enabler to allow interoperability between the defendant's programs and the plaintiff's devices. Further the court ruled that companies cannot use DMCA in conjunction with copyright law to create monopolies of manufacturer goods for themselves.

3. Whether, and to what extent, innovative services are being enabled and/or frustrated by the application of existing copyright law to software in everyday products;

While the ownership of the copyrighted software rests with the author(s) of such copyrighted work, copyright law also allows/would allow the owner of the program to make another copy or adaptation for use on a computer or to allow circumvention of a technological measure for indicative service related scenarios as indicated below.

Permitting the reproduction of computer programs for the purposes of machine maintenance or repair has been shown to be enabled by copyright law, as evidenced by various copyright related case laws. This clearly includes transformation of computer hardware/firmware of a consumer product enabled by software to extend the capabilities of the product or for modifying it to repurpose it for uses other than its originally intended use by the maker or to offer ancillary services (e.g. research and services in diagnosis and resolution of security/encryption technologies) around software-enabled consumer products. Overall, this will boost innovative services to be offered by organizations in addition to that provided by the product manufacturers themselves and inject further innovation in the overall consumer product and/or service ecosystem.

In many areas such as automotive, consumer electronics, medical devices etc., the boost to innovative services that may be offered by the ancillary service organizations (e.g. investigation/diagnosis and resolution services in various industries) would have a positive impact in creating a competitive services ecosystem for a software-enabled consumer product, and a lowered total cost of ownership of the same.

All in all, existing US copyright law enables the realm of innovative services in the context of software in everyday products. As in the case of products, it is recommended to have as much clarity as possible in the US copyright law to cater to different categories of innovative services.

4. Whether, and to what extent, legitimate interests or business models for copyright owners and users could be undermined or improved by changes to the copyright law in this area

Business models for consumer products and services are evolving at a rapid rate. The production and supply chain for a software-enabled consumer product is becoming increasingly global in nature. A consumer product today (manufactured in the US or whose manufacturing may be outsourced to any other country) may have a large number of software and hardware components sourced from different parts of the world, each with their own copyright laws (that may differ from US copyright law) and business practices (relating to treatment of aspects discussed in this study).

Also, with the increasing importance of cloud based products/services, a consumer product's functionality will increasingly be dependent upon, controlled by or extended in a significant way by software on the cloud. This also will lead to national boundaries being crossed in the matter of rights of copyright owners and users and also the rights of any third parties making interoperable products or services (either imported through physical boundaries or online connections).

It is therefore important that any change/advancements in US copyright law considers implications of this globally connected supply chain of components and services.

The rights of copyright owners and manufacturers of software-enabled consumer products on one hand, consumers and users on the other, and third party interoperable product and service providers need to be balanced to lead to a more sustainable innovation ecosystem. It is recommended therefore that US copyright law and related guidelines be closely monitored to see if it can keep pace with the evolving business models and legitimate interests of all stakeholders in this arena.

5. Key issues in how the copyright law intersects with other areas of law in establishing how products that rely on software to function can be lawfully used.

From the various examples seen above, it is seen that the reverse engineering of consumer products, including the software enabling them, for purposes of creating transformative products or for enabling interoperability or for offering ancillary services should be explicitly enabled by copyright law.

These rights should not extend to enabling distribution, sale or lease of multiple copies of the original consumer product – a basic protection offered to copyright rights owner. Similarly any other IP rights that are associated with the original product such the patents, trademarks and trade secrets, need to continue to be honored. Beyond this, any limits on lawful use of the product should be governed by relevant laws on consumer protection, warranty, safety, environmental and other laws of the land.

Once this is ensured, it can be safely said that the objective of copyright law is indeed to promote the progress of science and useful arts (and in this context – the software-enabled consumer products and services) and to not hinder/limit the same.